Research scope and source map

Research date: 2026-10-22. **Status: lecturer overview inspected; teammate review pending.**

1. Lecturer-first scope check

The user downloaded `source-materials/chap6 Behaviorist AI.pptx` after the browser download was blocked. Inspection found two slides: a title and an overview. See the scope review for their exact contents and the coverage map.

Slide 2 explicitly includes RL basics, algorithms and techniques, MDPs, robotics and games, applications, and limitations. The main chapter now develops RL and an MDP rather than treating learning as a short optional comparison. The detailed fixed-rule controller is retained as a supplementary comparison.

The slides do not define Behaviorist AI, name an algorithm, or establish depth, duration or assessment expectations. Agents, reactive priorities, finite-state control and subsumption remain supporting concepts selected for this resource. Q-learning is our proposed beginner algorithm, not a lecturer requirement. The separate Shared Assignment Rules attachment remains unchecked.

2. Claim-to-source table

Locators use section titles where available because PDF viewer numbers, draft pagination, and published-edition pages can differ. Sources support the concepts, not this lesson's invented thresholds, corridor, or teaching schedule.

ID	Claim supported	Source and exact locator	Verification and limits
S1	Agents perceive through sensors and affect their environment through actuators; simple reflex behaviour uses the current percept, while other agent designs retain internal state.	Russell & Norvig, *Artificial Intelligence: A Modern Approach*, author-hosted Chapter 2 sample, §2.1 "Agents and Environments," Fig. 2.1; §2.5 summary bullet beginning "Simple reflex agents."	Relevant search-indexed text inspected. The sample's displayed p. 33 for §2.1 differs from the Global Edition's pagination; do not mix them. Not a review of the whole textbook.
S2	A controller receives percepts and issues commands; sensors/actuators can be unreliable and actual action can differ from the command.	Poole & Mackworth, *Artificial Intelligence: Foundations of Computational Agents*, 3rd ed., author-hosted §2.1, opening paragraphs, Fig. 2.1, and §2.1.1 "Controllers."	Relevant author-hosted indexed text inspected. Supports keeping the toy model's ideal sensors distinct from real hardware.
S3	Behaviour-producing layers can connect sensing directly to action; Brooks proposed incrementally adding interacting layers.	Brooks (1991), *Intelligence without representation*, §4.2 "Decomposition by activity."	Full text available through web extraction; relevant section inspected. A historical research approach,

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			not a universal definition of AI.
S4	Brooks's subsumption implementation uses asynchronous finite-state modules and suppression/inhibition connections, rather than merely one ordered conditional chain.	Same Brooks (1991) paper, §6.2 “An instantiation of the methodology,” paragraphs beginning “Each layer in the subsumption architecture” and “Layers are combined,” Fig. 2.	Relevant full-text passages inspected. For this host's PDF these appear around viewer pages 7–8. Use section and paragraph locators if comparing another layout.
S5	The original layered architecture builds levels of competence from asynchronous communicating modules while lower layers continue operating.	Brooks (1985), MIT AI Memo 864, *A Robust Layered Control System for a Mobile Robot*, abstract; author-hosted PDF.	MIT repository abstract inspected. PDF loaded but produced no extractable text; abstract supports only this limited claim. Memo dated September 1985; distinguish it from the 1986 journal publication.
S6	Finite-state robot controllers have discrete modes, an initial mode, transition conditions, and tie-breaking when several transitions apply.	Hauser, author-hosted *Robotic Systems*, Robot Control, subsection “State machines,” Fig. 6.	Relevant author-hosted indexed text inspected. This chapter's transition-before-action convention is our explicit design choice, not a convention imposed by the source.
S7	State machines can specify robot programs using sensor readings as inputs and control commands as outputs.	MIT 6.01, Spring 2011 course textbook, Chapter 4, printed p. 118, numbered item 1 “Synthetically.”	Relevant indexed textbook extract inspected. No claim that a robot lab was performed.
S8	RL learns from interaction to improve accumulated reward, rather than simply executing an unchanged controller.	Poole & Mackworth, author-hosted Chapter 13 “Reinforcement Learning”, introductory paragraph.	Relevant indexed introduction inspected. Its fully observable single-agent scope is narrower than all RL.
S9	Exploration gathers experience; exploitation selects actions favoured by current estimates.	Poole & Mackworth, author-hosted §13.5 “Exploration and Exploitation”, opening two bullets and following discussion.	Relevant indexed section inspected. The chapter avoids convergence guarantees and advanced strategy equations.
S10	Sutton and Barto discuss trial-and-error interaction and the exploration/exploitation tradeoff.	Sutton & Barto, author-hosted March 2018 draft, §1.1 “Reinforcement	Author-hosted indexed §1.1 passage and contents inspected; direct full-file

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		Learning," printed p. 3. §1.3 "Elements of Reinforcement Learning" begins on printed p. 6 in this draft's contents.	retrieval timed out. Do not claim §1.3 body was fully inspected. Use S8/S9 for this draft's main RL explanation.
S11	An agent can explore transitions/rewards and use feedback to estimate a policy.	UC Berkeley CS188, official §5.1 "Reinforcement Learning", introductory contrast between offline planning and online learning.	Relevant indexed university text inspected. The no-prior-model setup in this introduction is an example setting, not a necessary property of every RL system.
S12	The lecturer's overview explicitly includes RL basics, algorithms, MDPs, robotics and games, applications and limitations.	Supplied Chapter 6 deck, slide 2 "Overview"; slide 1 is the title.	Both slides inspected from the local PPTX package, including hidden-slide flags and linked notes/pictures. The deck contains no detailed teaching content.
S13	An MDP specifies states, actions, transitions and rewards; the Markov assumption concerns the sufficiency of the current state; discounting weights later rewards.	UC Berkeley CS188, official §4.1 Markov Decision Processes, opening specification, §4.1.1 "Finite Horizons and Discounting," §4.1.2 "Markovianess."	Relevant university text and equations retrieved. The vacuum state, rewards and gamma are original design choices.
S14	Sampled tabular Q-learning updates an action estimate toward reward plus discounted maximum next-state Q-value.	UC Berkeley CS188, official §5.3.3 Q-Learning, "sample" and moving-average equations.	Relevant university text/equations retrieved. The six updates are independently calculated for our invented task; no convergence claim.

3. Original teaching design versus sourced facts

Sourced concepts: sensor/actuator interaction, reflex versus remembered state, behaviour layers, finite-state transitions, reward-driven learning, exploration/exploitation.

Original teaching proposals: the four-tile fixed controller, two-tile learning task, energy thresholds, rewards, action costs, mode guards, scripted updates, questions/answers, sequence and SVG layouts. They are invented instructional models, not experiments or claims about a commercial vacuum. AIMA also uses vacuum worlds; this corridor is a separately specified teaching example, not a reproduced figure or exercise.

Design deductions: the return-energy bound, oscillation counterexample, and reward-example conclusions follow from our stated assumptions. Check them against the model rather than attributing them to Brooks or BIT.

4. Research limits and next checks

This is a focused primary-source research draft, not an exhaustive literature review. Several websites were available only through indexed excerpts; those limits are recorded above. No third-party summaries are used as factual authorities. Retain exact edition/section names when checking the cited text. Do not redistribute entire lecturer decks or textbooks in the public folder without appropriate permission.

The lecturer overview has now been checked, but its expected depth remains unspecified. Next checks are another member's substantive review, beginner accessibility and the actual rendered PDF/assets when the team packages Stage 1. The user will complete personal logs and significant AI-use records separately.